

Identifying the Best Defensive First Baseman in Baseball

A Computer Vision and Data-Driven Proposal

Casey Hackett

August 28, 2025

Overview

This project aims to design a set of models that quantify and rank defensive performance at first base. Unlike other infield positions, first basemen are evaluated not only on range but also on their ability to receive throws, handle scoops, and make situational decisions. By combining Statcast data, computer vision techniques, and probabilistic modeling, this project seeks to identify the best defensive first baseman in Major League Baseball and provide actionable insights for roster construction and player development.

Motivation

First base defense has a significant impact on run prevention. While traditional metrics such as fielding percentage and errors offer limited insights, modern data collection enables a granular evaluation of:

- **Range and reaction time** on ground balls and line drives.
- **Receiving ability**, including scoops, stretches, and handling bad throws.
- **Decision-making**, such as bunt defense and coverage choices.

Developing robust models for these components provides teams with a competitive edge in player evaluation, coaching, and in-game strategy.

Methodology

The project will consist of four integrated models:

Model 1: Range and Reaction Model: A spatial probability model estimating the likelihood of a first baseman fielding a batted ball given its exit velocity, launch angle, spray direction, and player positioning. Outcomes compared to league-average probabilities yield runs saved estimates.

Model 2: Receiving and Scoop Model: A computer vision pipeline (YOLOv8 + pose estimation) to track throws arriving at first base. Logistic regression or neural classifiers will estimate the probability of a successful scoop or stretch given throw trajectory, with deviations from league averages quantified in runs.

Model 3: Decision-Making Model: A Markov Decision Process (MDP) framework evaluating situational choices (fielding vs bag coverage, bunt defense). Each choice will be assigned an expected run value relative to optimal decisions.

Model 4: Aggregate Defensive Value Model: A weighted combination of the above outputs into a unified metric (Defensive Runs Saved at First Base), normalized across the league for comparability.

Expected Deliverables

- Visual demos of plays with pose estimation overlays (scoops, stretches).
- Probability maps of expected vs actual fielding performance.
- Statistical rankings of MLB first basemen by defensive contribution.

Applications to Decision-Making

- **Front Office:** Inform free agent signings and trade valuations by quantifying defensive contributions in runs saved.
- **Player Development:** Diagnose weaknesses (e.g., scooping vs range) and target individualized defensive training.
- **In-Game Strategy:** Provide coaching staff with optimal defensive positioning and bunt defense alignments.
- **Roster Construction:** Quantify trade-offs between offensive production and defensive value at first base.

Future Extensions

- Extend to other positions.
- Incorporate biomechanical injury-risk markers (fatigue detection via pose estimation).
- Adapt for real-time decision support during games.

Conclusion

This proposal outlines a framework for evaluating first base defense through a blend of probabilistic modeling and computer vision. By breaking down defense into its core components and validating against Statcast data, the project offers a rigorous approach to answering the question: *Who is the best defensive first baseman in baseball?*